MCO-1D

Software Reference — Rev 22

Ground station command reference for commissioning engineers. For pilot operation see the Operator Manual v22.

Document	MCO-1D_Software_Reference_v22
Revision	22 — August 2026
New in v22	Wing coordination commands · Speed/climb/aerobatic diagnostics · Spatial awareness · Gap negotiation
Audience	Commissioning engineers — not required for pilot operation

Architecture Overview

The MCO-1D flight software runs on the Raspberry Pi aboard the aircraft. The ground station (GS) connects via WebSocket and sends commands. The AI assistant (Claude) translates pilot natural language into GS commands.

LAYER	WHAT IT DOES
mco_wave_library.py	All flight control, learning, and intent classification. Runs on the Pi. 67,700+ lines.
mco_ws_bridge.py	WebSocket server on the Pi. Accepts GS commands, returns telemetry and acks.
mco_api_proxy.py	Proxy on the laptop. Connects browser to Anthropic API.
mco_ai_ground_station.jsx	Browser interface. Six steam gauges + conversation. AI translates pilot speech to GS commands.

Control loop

50 Hz main loop: pilot_intent_tick → CIC (climb) → MIC (aerobatics) → wing_guard_tick → spatial_awareness_tick → ODrive CAN output. Discovery and genome updates run at 1 Hz during cruise.

Arm / Disarm / Status

`arm`
Arm the aircraft. Requires prior preflight check.

`disarm`
Disarm (ground only unless in emergency).

`soft_stop_status`
All axis status summary.

`fault_summary`
Read and decode active faults.

`mechanism_weight_report`
Show learned mechanism weights per axis.

Wing Coordination — NEW v22

- NEW `wc_set_fuselage_width [mm]`
  Set measured spar-to-spar fuselage width. REQUIRED before TC-01. Calibrates contact geometry, rigid/TPU zone boundaries, amplitude limits.
- NEW `wc_set_tpu_start [mm]`
  Set measured TPU winglet boundary from root pivot. REQUIRED before TC-01. Determines which contact zone is rigid (hard stop) vs TPU (clap allowed). Optionally: wc_set_tpu_start [mm] [pair_0-3] for per-pair values.
- NEW `wc_geometry_report`
  Full two-zone geometry report. Shows contact radius at each amplitude, rigid vs TPU zone classification, amplitude limits. Confirm 96° contact falls in TPU zone for all four pairs.
- NEW `wc_fuselage_guidance`
  Print design guidance for fuselage width selection. Shows biological reference points and optimal target range.
- NEW `clap_enable [dorsal|ventral]`
  Open clap gate. Allows validated wing contact in TPU zone at stroke reversal. Amplitude increases to 100.5°. Guard still validates timing every tick.
- NEW `clap_disable`
  Close clap gate. Return to non-contact operation.
- NEW `clap_status`
  Two-zone amplitude limits, tip separations, clap count.
- NEW `wing_guard_status`
  Recent guard interventions and severity.

Attitude Hold / Hover

`attitude_hold_status`
PID gains, trim state, attitude errors.

`hover_mode_status`
Hover controller state, blend fraction.

`position_hold_engage`
Engage GPS position hold.

`position_hold_disengage`
Release position hold.

Speed Performance — NEW v22

- NEW `speed_performance_status`
  Speed regime boundaries, Strouhal numbers at max capability, active speed foci and hypotheses.

Climb Performance — NEW v22

- **NEW** `climb_performance_status`
  Both climb optima (ROC and angle), target speeds, configurations seeded, ALSS curriculum coverage.
- **NEW** `climb_intent_status`
  Diagnostic: current CIC mode, blend fraction, thresholds. Not surfaced to pilot during normal flight.

Aerobatic Flight — NEW v22

- **NEW** `manoeuvre_intent_status`
  Diagnostic: current MIC mode, urgency, per-manoeuvre dwell/blend times. Not surfaced to pilot during normal flight.

Focus / Learning

`focus [name]`
Set mission focus. See focus list below.

`focus_clear`
Clear active focus.

`discovery_master_on`
Enable discovery during current flight.

`discovery_master_off`
Disable discovery (hold current config).

`persist_checkpoint`
Save all learning to disk immediately.

`gs_inject_note [text]`
Inject a text note into the flight log.

Focus names (complete list v22)

FOCUS	DESCRIPTION
`gust_tolerance`	Wind and turbulence stability
`position_hold`	GPS position hold accuracy
`smooth_flight`	Minimum vibration / camera payload
`crosswind_landing`	Crosswind approach and landing
`endurance`	Minimum power consumption
`sprint`	High-speed sprint with safety margin
`sar_hover`	SAR hover and confined-space ops [persistent]
`storm_resilience`	Maximum dihedral, load shedding [persistent]
`aerobatics`	Roll / pitch / yaw authority and precision
`kestrel_hover`	Thermal soaring mode [persistent]
`top_speed`	Maximum speed pursuit (28 m/s Vne) [NEW v22]
`fast_cruise`	Efficient high-speed transit [NEW v22]
`best_climb_rate`	Maximum rate of climb at Vy [NEW v22]
`best_climb_angle`	Maximum climb angle at Vx [NEW v22]

ALSS

`alss_set_fence [lat] [lon] [radius_m] [height_m]`
Set practice fence.

`alss_start`
Start autonomous learning sortie programme.

`alss_abort`
Abort current sortie, return home.

`alss_status`
Sortie number, class, current segment, scores.

Ceiling Exploration

`ceiling_explore_arm`
Step 1: print checklist, arm for 60s window.

`ceiling_explore_confirm`
Step 2: begin probe (within 60s of arm).

`ceiling_explore_abort`
Descend immediately, end probe.

`ceiling_explore_status`
Current altitude, saturation margins, LEV risk.

Spatial Awareness — NEW v22

- NEW `spatial_awareness_enable`
  Enable obstacle sensing ring.
- NEW `spatial_awareness_disable`
  Disable obstacle sensing.
- NEW `obstacle_status`
  Live obstacle map from six ToF sensors.
- NEW `vio_status`
  VIO position estimate and optical flow velocity.
- NEW `gap_set_width [m]`
  Set gap width in metres. Selects strategy: >0.55m flapping / 0.20-0.55m sideways / 0.12-0.20m ballistic.
- NEW `gap_enter`
  Begin gap negotiation passage.
- NEW `gap_abort`
  Abort passage, recover to normal flight.
- NEW `gap_status`
  Current gap mode, phase, tip separation.
- NEW `canopy_approach [lat] [lon] [alt_m]`
  Accipiter flap-flap-glide canopy approach to target.
- NEW `perch_land`
  Vertical descent to surface with position hold.
- NEW `confined_exit [height_m]`
  Pop-up vertical climb from confined space. Front wings only until clear height.

Lever Brake Modes

`jhook_engage`
Engage J-hook post-stall perch (lever 93-97%, >20m).

`jhook_abort`
: Abort J-hook, recover to normal.

`brake_status`
: Current lever brake mode and parameters.

Telemetry Keys (selected)

`V_ms`
: Pitot airspeed (m/s)

`h_agl_m`
: Altitude above ground (m)

`vel_down_ms`
: Vertical velocity (m/s, negative=climbing)

`roll_rate_dps`
: Roll rate (deg/s)

`pitch_rate_dps`
: Pitch rate (deg/s)

`motor_I_A`
: Motor current (A)

`bus_voltage`
: Battery voltage (V)

`fault_count`
: Active fault count

- `NEW wc_min_sep_m`
  Minimum wing tip separation all pairs (m) [NEW]
- `NEW wc_clap_count`
  Validated clap events this session [NEW]
- `NEW cic_mode`
  Climb intent: sustained/best_rate/best_angle/zoom [NEW]
- `NEW mic_manoeuvre`
  Aerobatic manoeuvre currently classified [NEW]
- `NEW mic_urgency`
  Stick urgency scalar (0–max) [NEW]
- `NEW spa_threat_m`
  Nearest obstacle distance (m) [NEW]
- `NEW spa_fwd_m`
  Forward sensor reading (m) [NEW]

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